بسم الله الرحمن الرحيم

Surface anatomy of permanent anterior teeth

Anterior teeth

They include:

Incisors

And

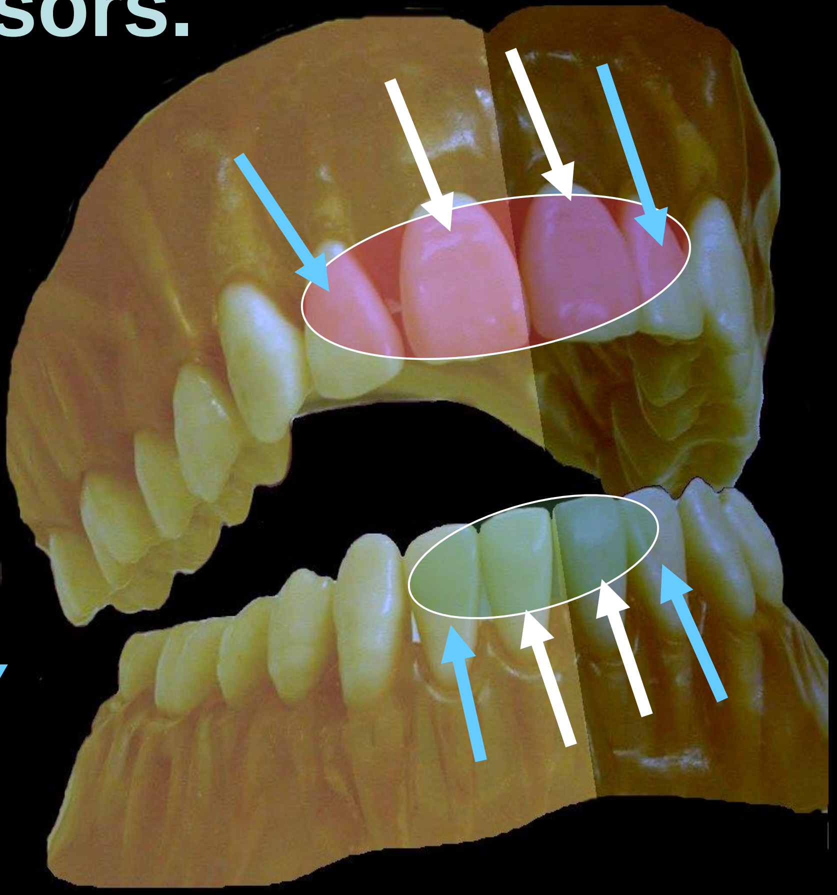
Canines

Incisors

There are four maxillary incisors

and four mandibular incisors.

**** Two central incisors contact with each other in the midline (mesially) and with the lateral incisors distally.***



**** Two lateral incisors contact with the central incisors mesially and with the canines distally.***

Number of lobes:

All anterior teeth (incisors and canines) are formed of four lobes, three labially and one lingually.

Note:

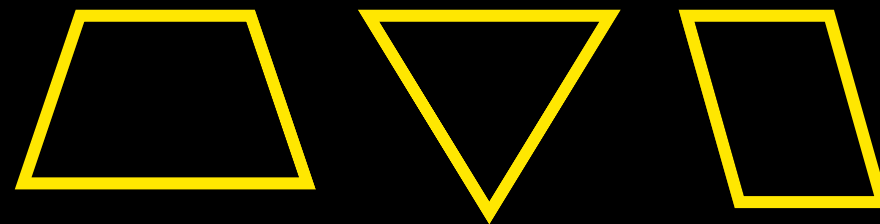
Mamelons in newly erupted incisors. Diastema between two upper incisors.

	Beginning of calc. (months)	Crown completed	Eruption	Root completed
$\dfrac{1\|}{1\|}$	3-4	- 3 years	7 **6**	+ 3 years
$\dfrac{2\|}{2\|}$	**10-12** 3-4		8 7	
$\dfrac{3\|}{3\|}$	4-5		**11** **9**	

For proper tooth description

We have to speak about :

- *The geometric outline of the crown.*



- *The outline form of the crown and root.*

- *The surface anatomy of the crown and root (anatomical landmarks).Then describe the root Number,shape and apex.*

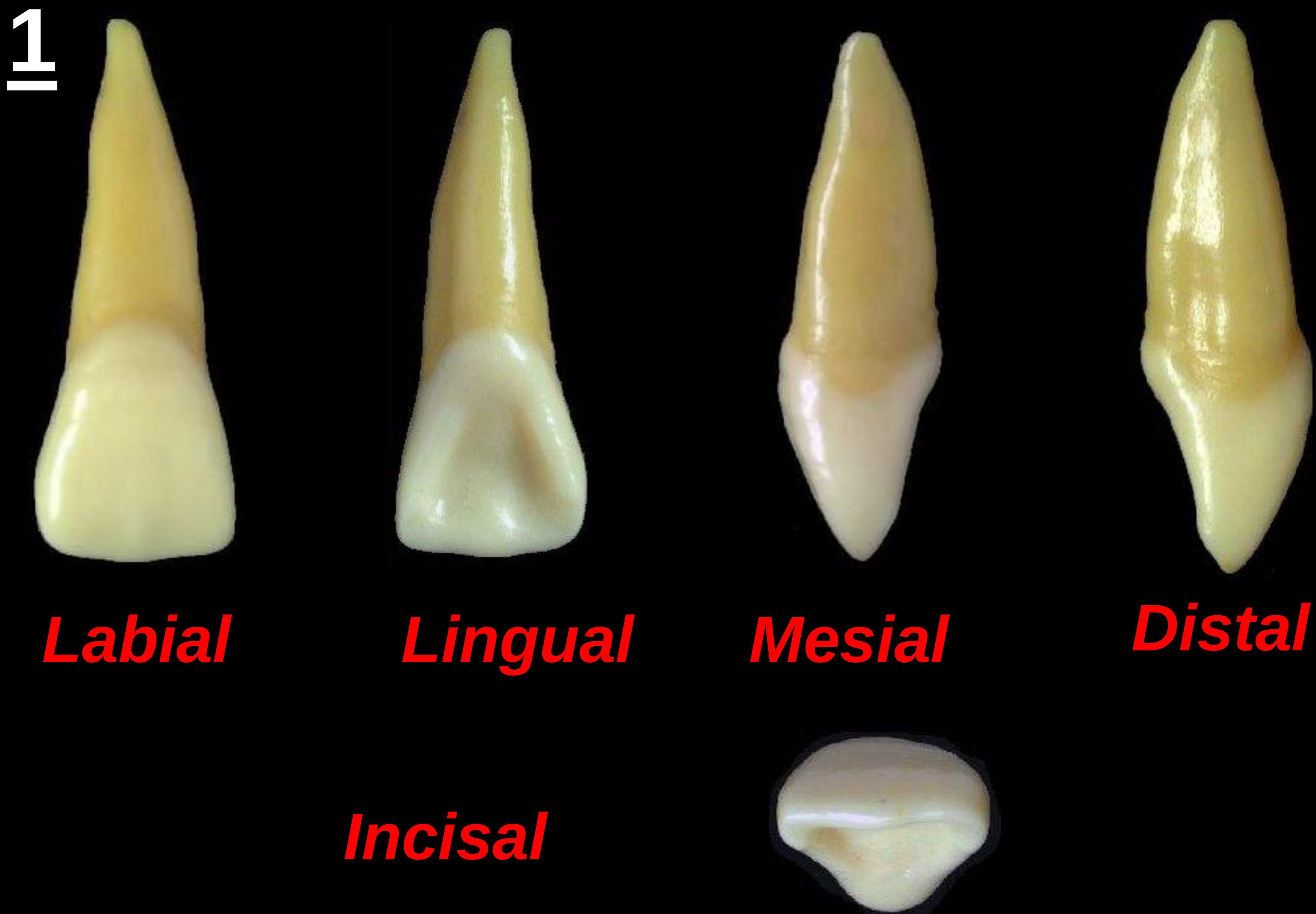
Permanent maxillary central incisor

1

Maxillary central incisor is the first tooth from the midline.

The maxillary central is the broadest and longest of all incisors.

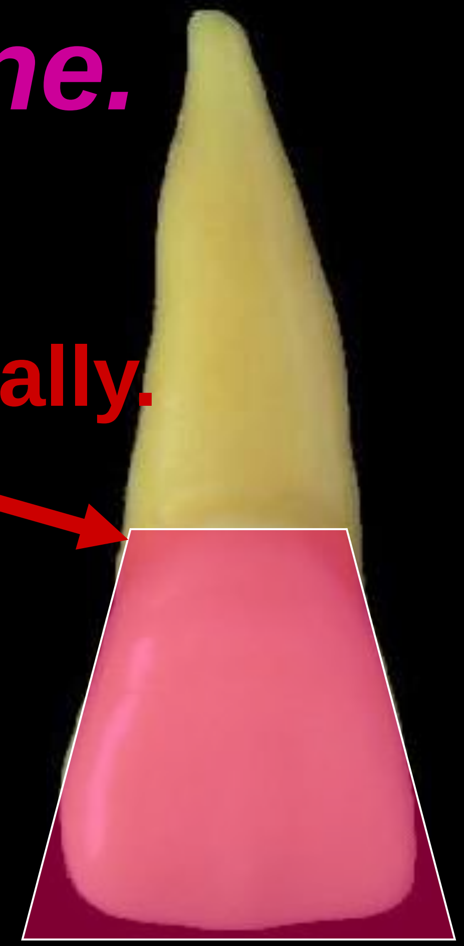
All teeth have 5 surfaces



Geometric outline of the crown

The labial and lingual surfaces have trapezoidal outline.

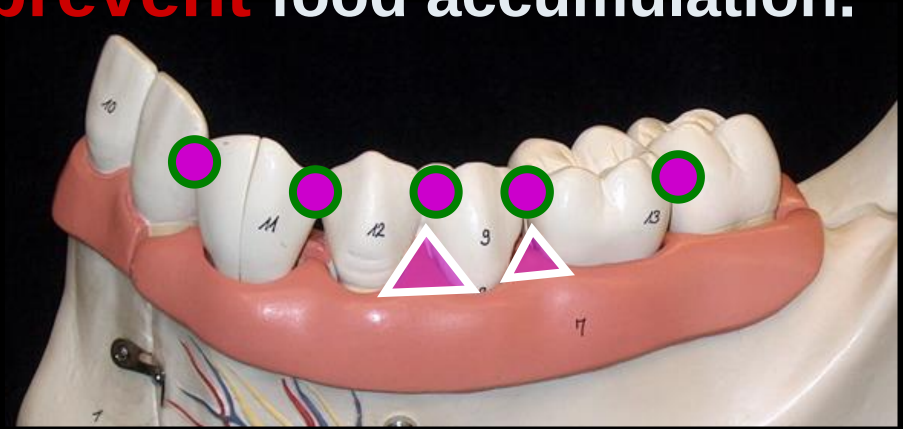
The smallest uneven side cervically.



Significance of the trapezoidal outline in protecting the periodontium.

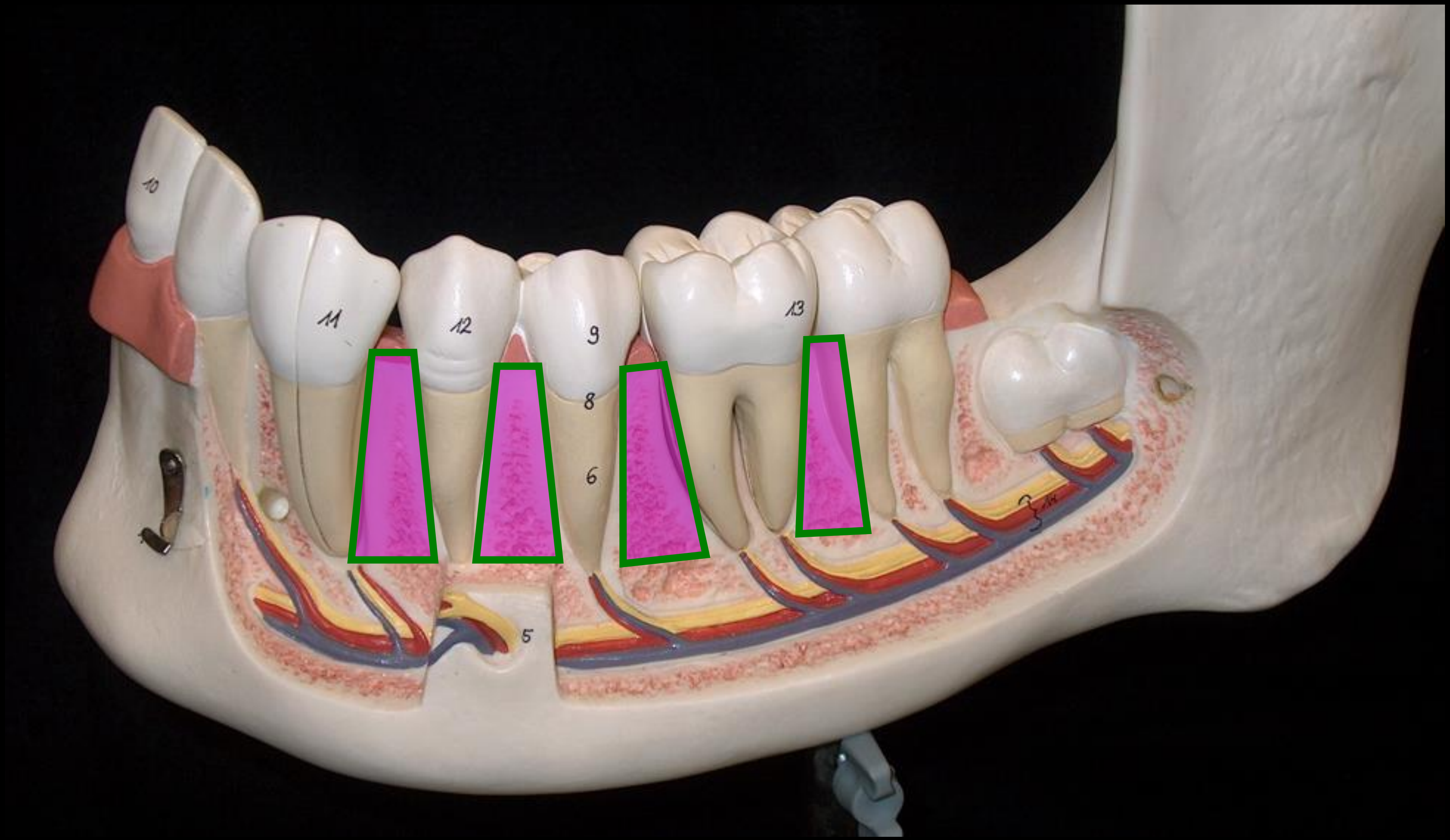
1- Provides contact between the teeth.

This gives stabilization of the dental arch, protect the inter-proximal soft tissue and prevent food accumulation.

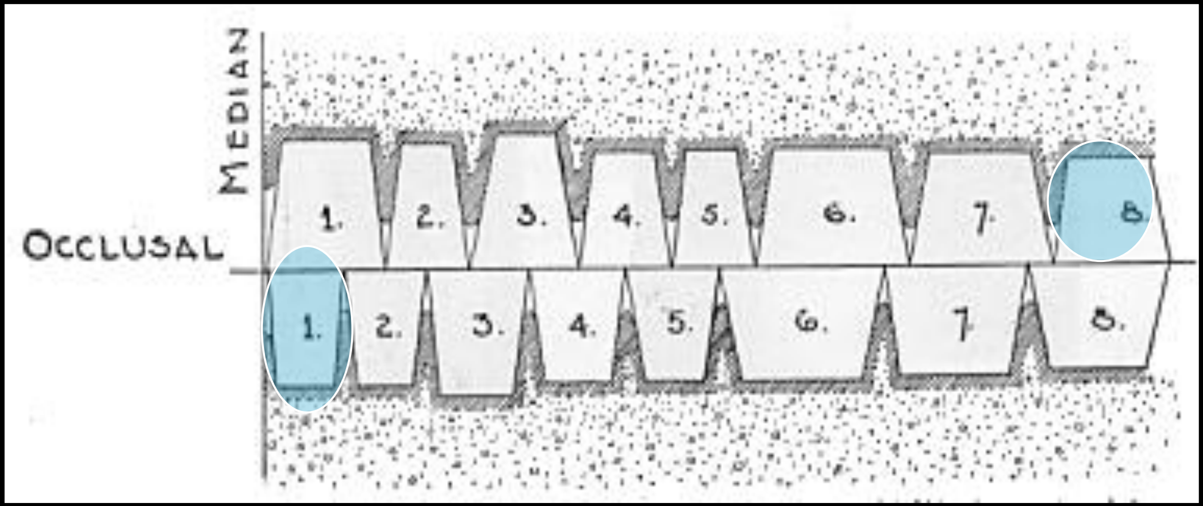


2- Provides inter-proximal spaces which contain inter-proximal gingival tissues.

3- Provides spacing between the roots of neighbouring teeth to allow sufficient supporting alveolar bone.

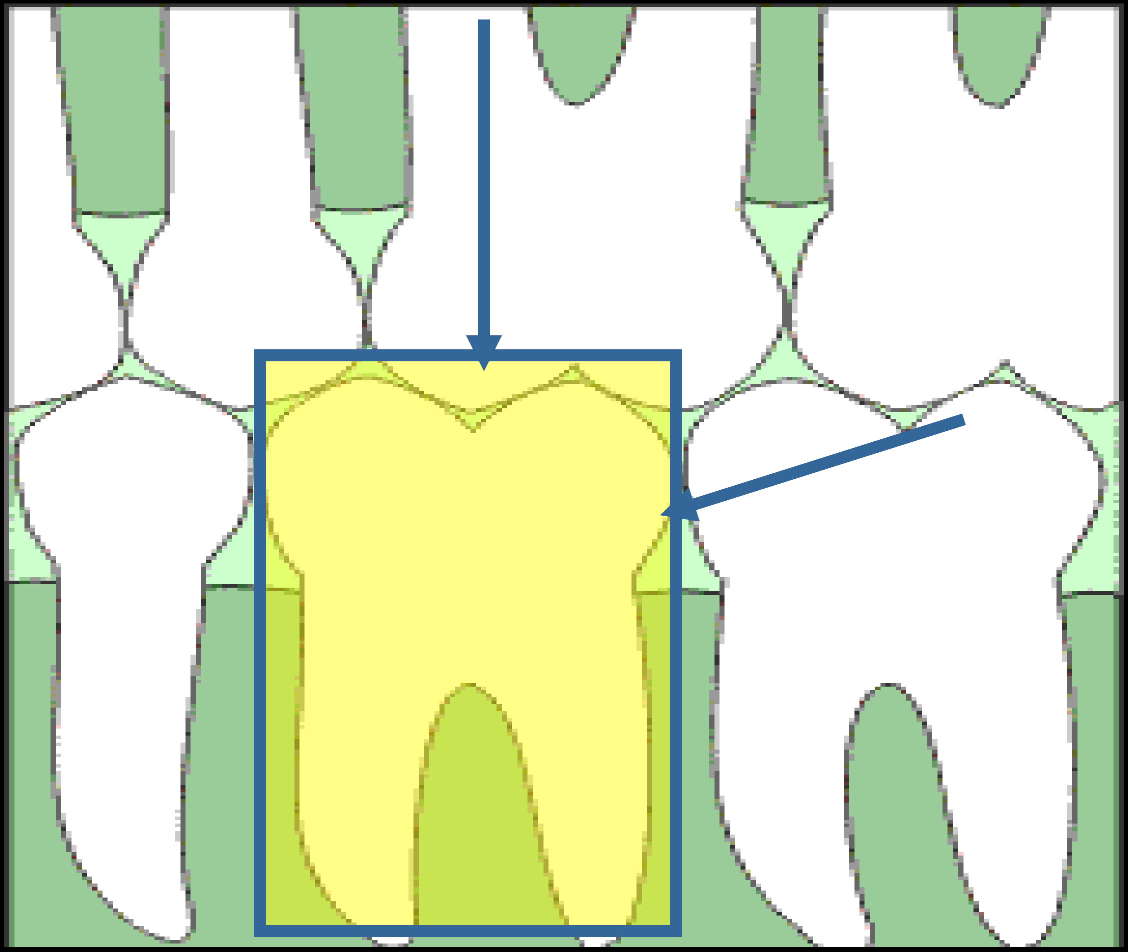


4- Allow each tooth in one dental arch to occlude with two opposing teeth except $\overline{1}$ & $\underline{8}$

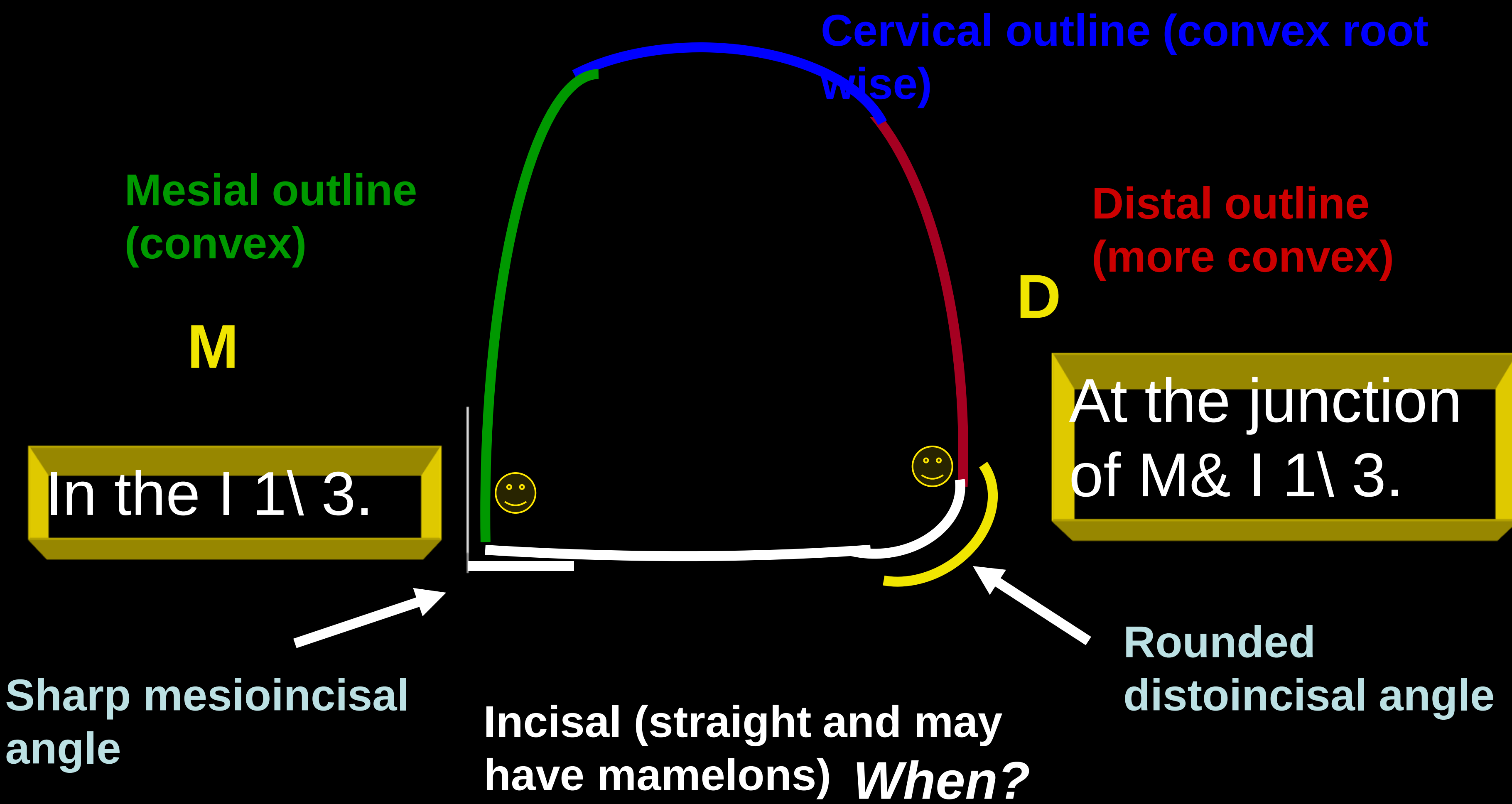


- **This arrangement distributes and reduces the occlusal forces exerted on the teeth.**

***This arrangement also prevents elongation of the antagonists and helps to stabilize the remaining teeth for a longer period than if the tooth has a single antagonist.**



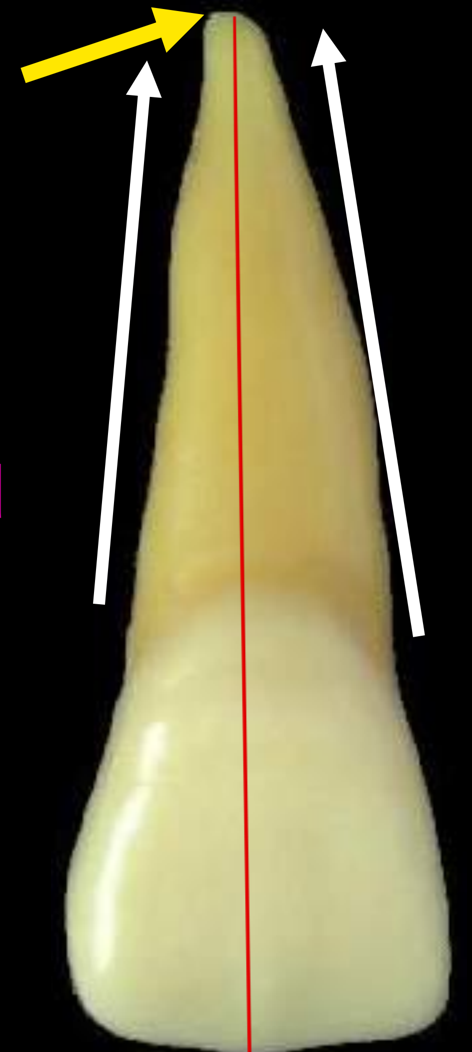
Labial and lingual outlines of the crown



Outlines of the root

- The mesial and distal outlines of the root are tapered to a blunt apex

- The apex is centralized on the long axis of the tooth ,so, extraction should be done by rotation movement



Surface anatomy of the crown and root.

Labial surface:

Elevations:

- The crown surface is smooth and convex with the maximum convexity at the cervical third (cervical ridge)

Depressions:

•Shallow developmental grooves could be seen separating 3 mamelons in newly erupted central incisor .

•The root surface is smooth and convex

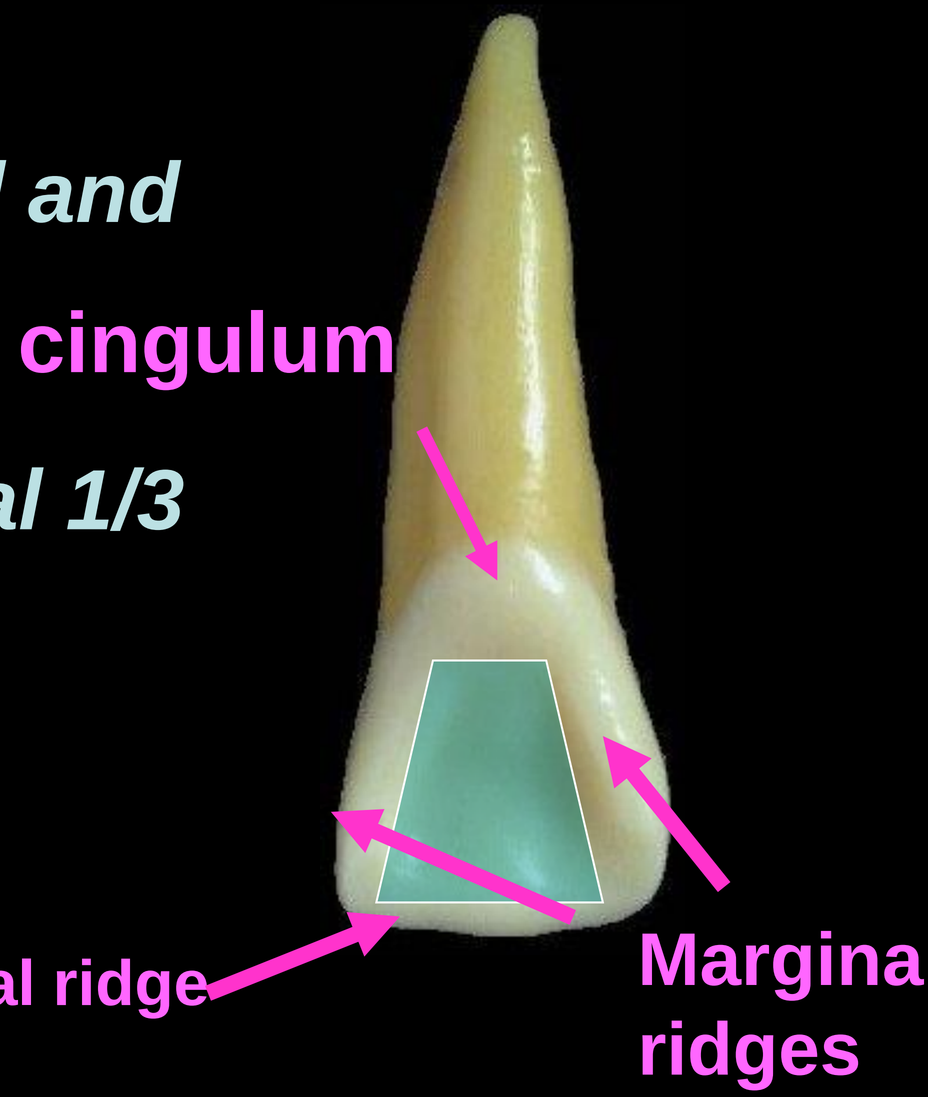
Lingual surface

Elevations:

- *Marginal ridges (mesial and distal)*
- *Cingulum at the cervical 1/3*
- *Incisal ridge*

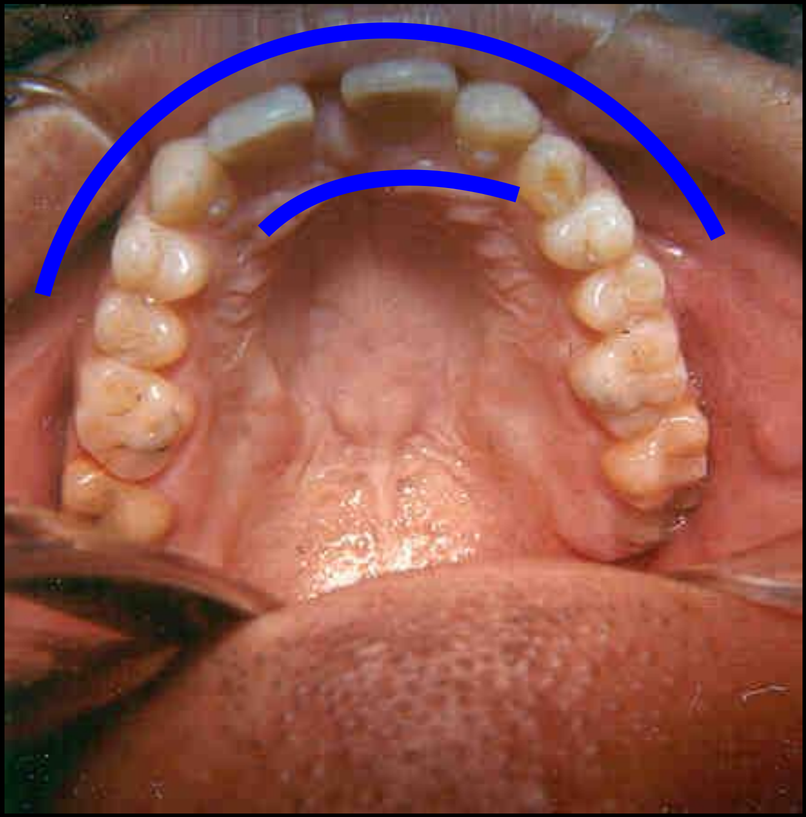
Depressions:

Lingual fossa



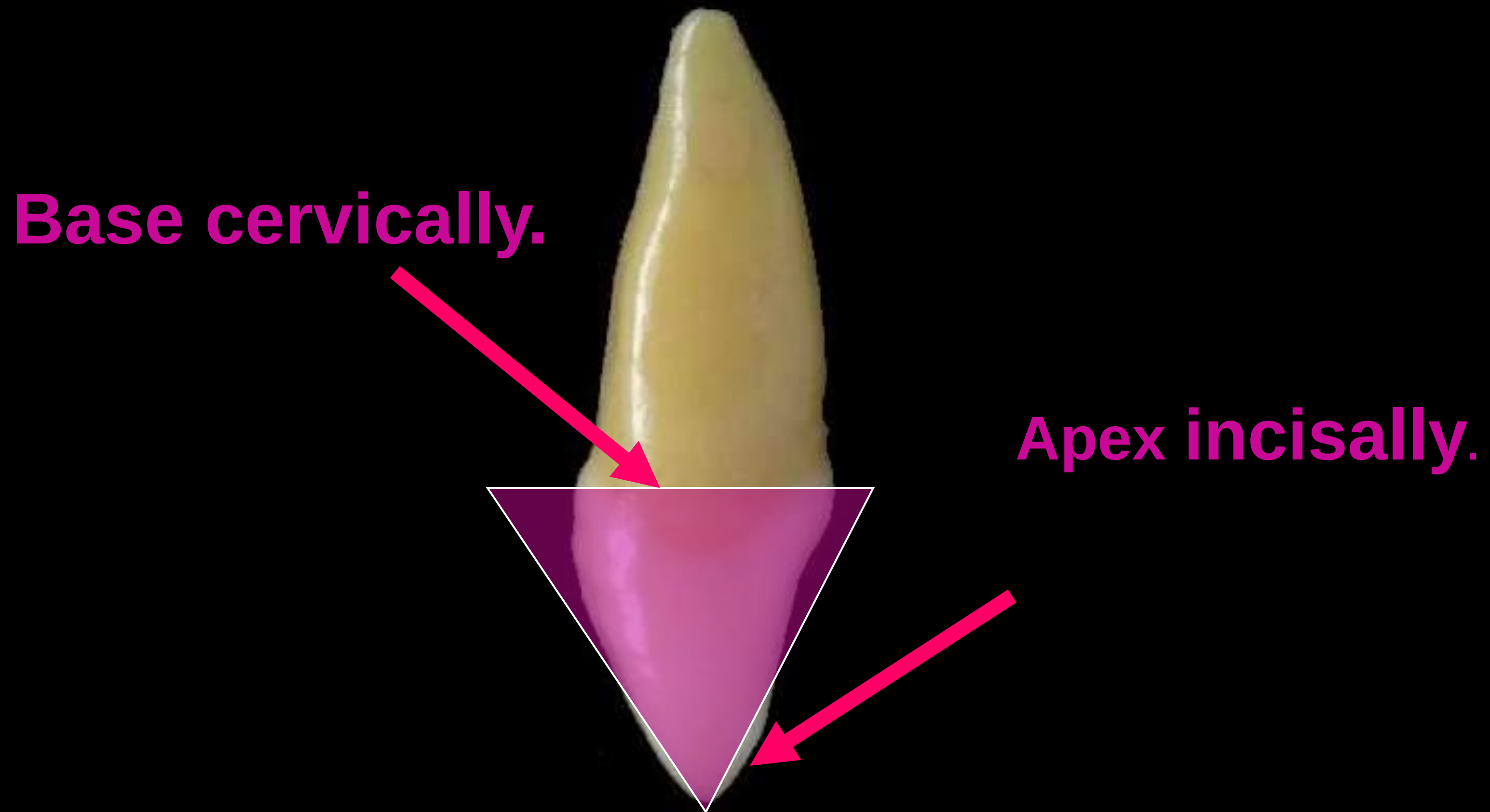
☺ Notice that in most of the teeth the lingual surfaces are narrower than the labial or buccal ones due to the lingual convergence.

This convergence of the teeth is to accommodate the larger arch size facially than lingually



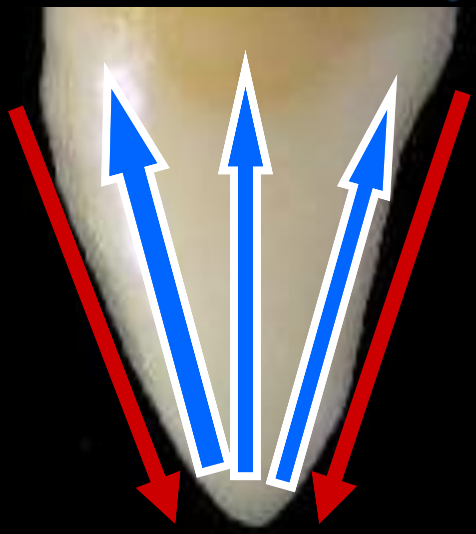
Geometric outline of the crown

Proximal (mesial and distal) surfaces have triangular outline



Significance of the triangular outline in protecting the periodontium.

1-It increases the teeth strength against masticatory forces.



2-It facilitates cutting through food materials.

3- This form allows the tooth to be self-cleansing.

Mesial and distal Outlines

It's Triangular and formed of

- ***Labial outline: convex with the maximum convexity at the cervical third which represent…….***
- ***Lingual outline:***

convex incisally which represent…….

Concave at the middle which represent…

Convex cervically which represent………

- ***Cervical outline: Curves incisally.***

Outlines of the root

The labial and lingual outlines are tapered from the cervical line to a blunt rounded apex

M

D

The crown has smooth convex proximal surfaces.

Contact areas:

Near the MI angle.

Near the junction of I & M 1\ 3.

Cervical line:

Curved incisally.

The curvature is shallower than mesially.

Root

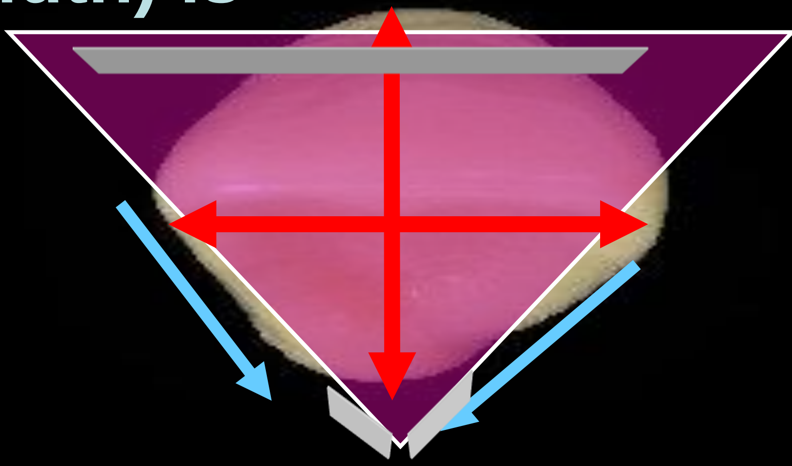
It has convex smooth surface.

Incisal aspect

Outline and surface anatomy

- The outline is triangular in shape.

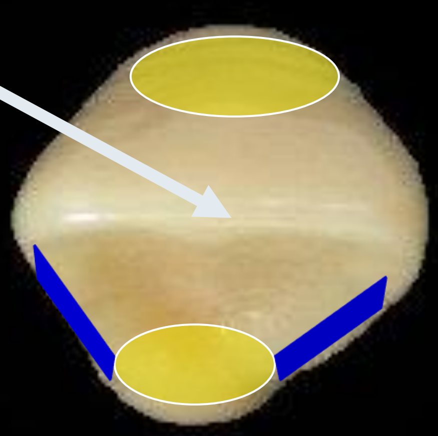
•The base is placed labially and the apex is lingually.
•The mesiodistal dimension (width) is greater than the labiolingual



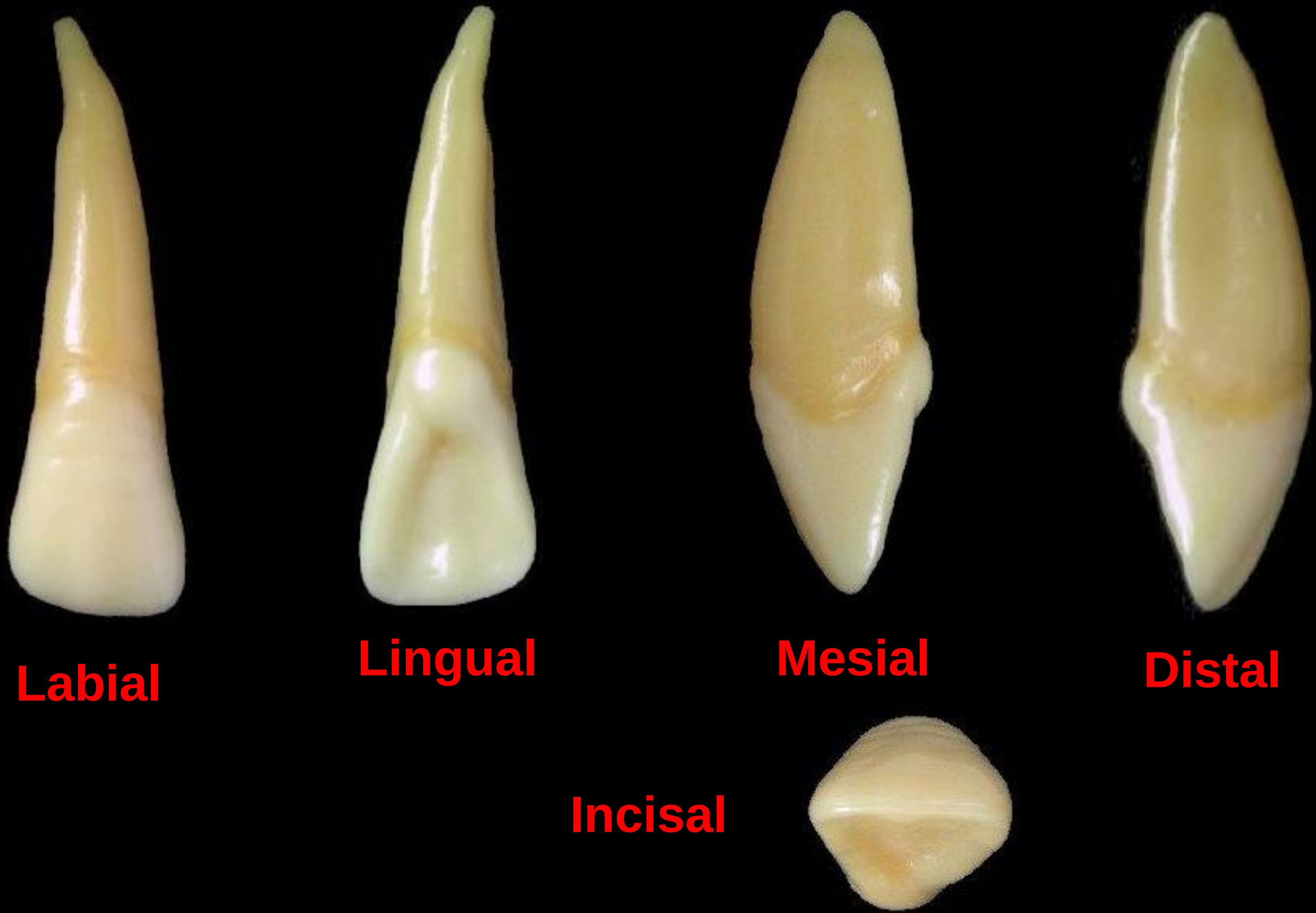
The crown is tapered lingually.

Note:

- The elevations and depressions in the crown appear in this aspect as the cervical ridge (labially) and the mesial and distal marginal ridges , incisal ridge and cingulum surrounding the lingual fossa (lingually).The incisal ridge is centralized labiolingually.



Maxillary lateral incisor



Number of lobes:

All anterior teeth (incisors and canines)are formed of four lobes, three labially and one lingually.

Note: minimum number of lobes in normal teeth is three, however peg-shaped 2 has two lobes.

The lateral incisor is ***smaller*** in all dimensions than the central incisor.

<u>1</u>

<u>2</u>

1

Labial surface

2

D  M

D  M

sharp M I angle.

rounded M I angle.

rounded D I angle.

more rounded D I angle.

Incisal outline Straight

Rounded

1

2

Newly erupted tooth has mamelons.

The mamelons are less pronounced.

Peg-shaped tooth is a form of 2 could be exist (two lobes, one labial and one lingual).

1

2

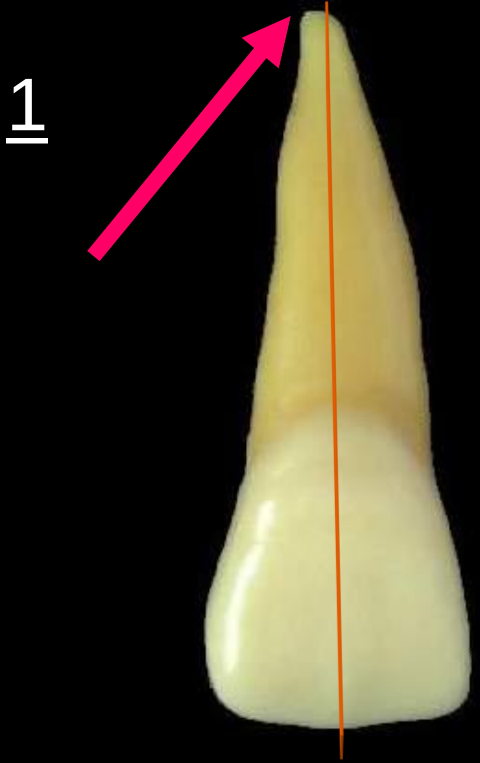
Labial surface:

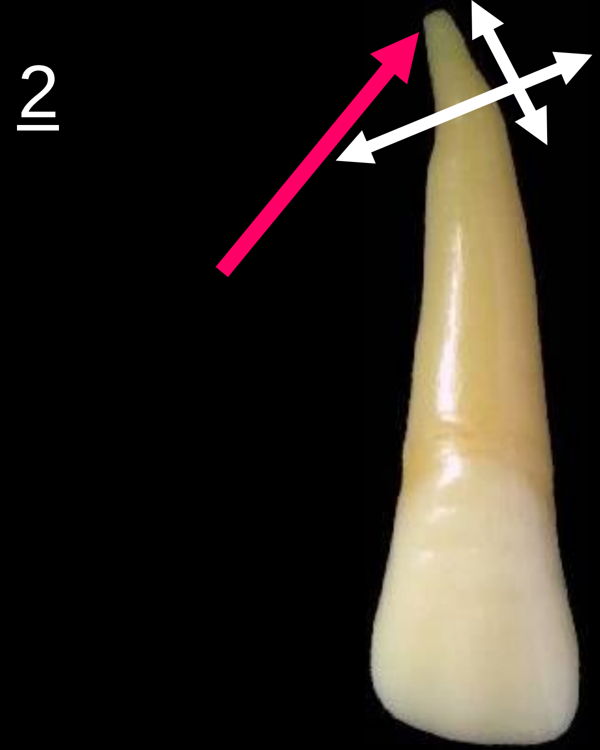
Convex or slightly flat.

More convex.

Labial D G.

Less prominent.





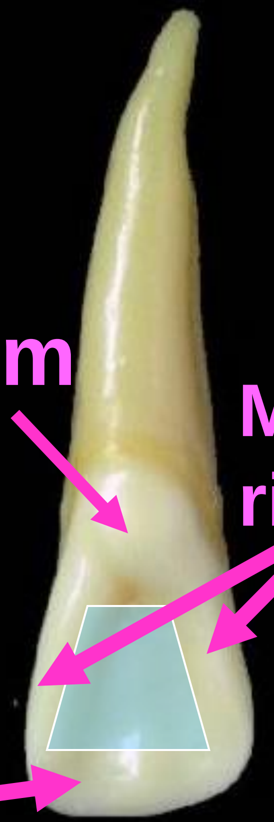
- **The mesial and distal outlines of the root taper to a blunt apex**
- **The mesial and distal outlines of the root taper to a pointed apex**

- **The apex is centralized on the long axis**
- **The apical 1/3 is inclined distally**

Lingual surface

Elevations:

- *Marginal ridges (mesial and distal)*
- *Cingulum (cervically)*
- *Incisal ridge*



Depressions:

Lingual fossa is more concave and circumscribed than that in <u>1</u>

Note:

- ***A lingual pit could be found in the <u>2</u> close to the cingulum. Notice that all elevations are well developed than those in <u>1</u>***

Mesial surface

1

Lb Lg

2

Lb Lg

The crown is long and thick labiolingually.

The mesial surface is flat

Contact area at I 1\3 near the M I angle.

The crown is shorter and thinner labiolingually

The surface is flatter.

Near or at the junction of I & M 1\3

1

Distal surface

2

C.A. at the junction of I & M 1\3

In the center of the crown.

Incisal aspect

1

2

Wide M D.

Labial and lingual O.L. are flat and broad.

Narrower in size.

Labial and lingual O.L. are more rounded.

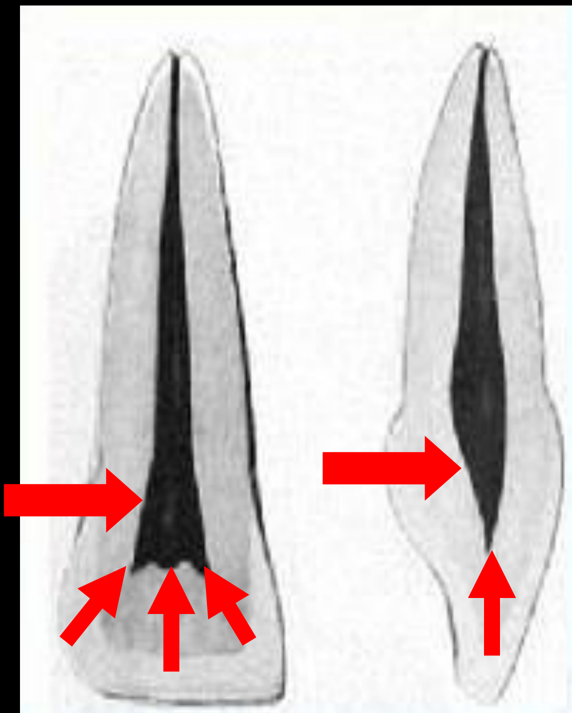
When palatal pit is present; it is located in the depth of the lingual fossa close to the cingulum .

Pulp cavity.

The pulp cavity is formed of:

***Pulp chamber* that is present in the crown. Its outline follows the outline of the crown.**

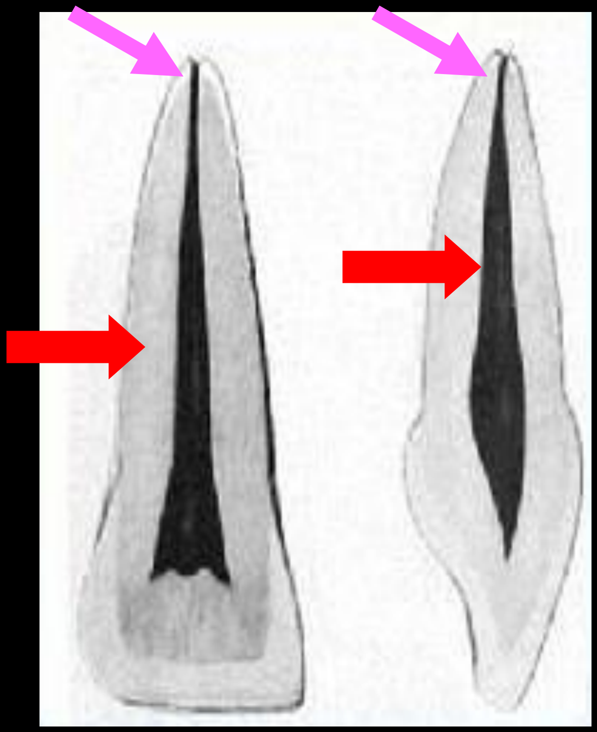
In young teeth, it has pulp horns related to each mamelon



2 Showed similar but smaller pulp cavity.

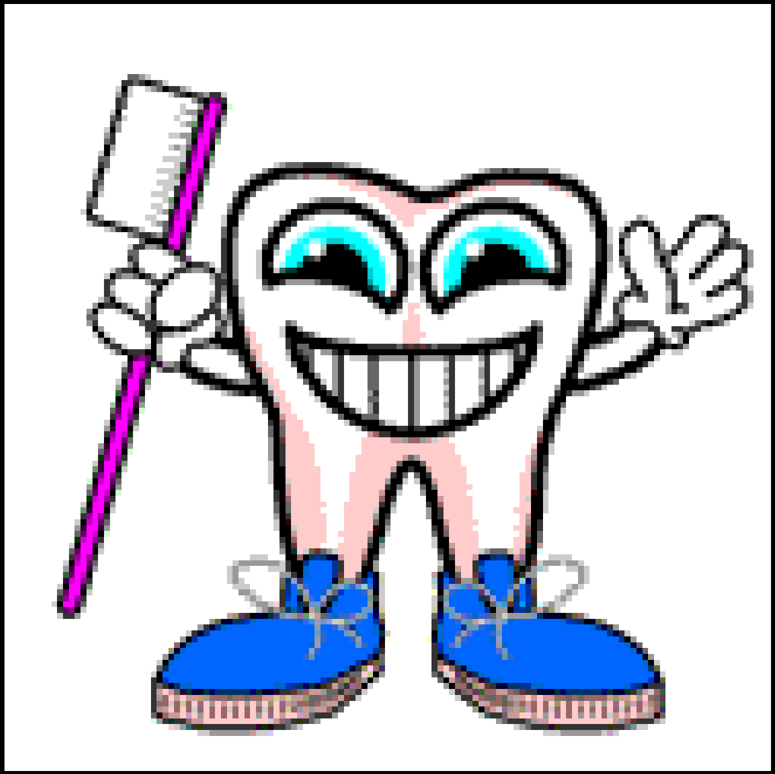
MD LL

The pulp chamber in 2 has one rounded or two less sharp pulp horns (M & D).

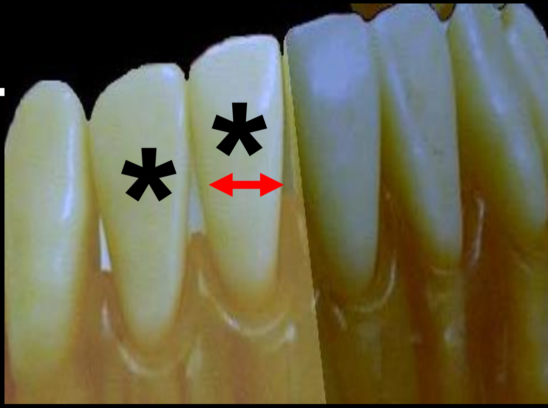


Root canal is present in the root and follows its outline. The number of root canals in the incisors is only one. The root canal ends in an apical foramen

Thank you



Mandibular incisors



*They are smaller than maxillary incisors.

*$\overline{1}$ is smaller than $\overline{2}$ which is the reverse

Of the situation in $\underline{1}$ & $\underline{2}$.

*The width is smaller than the thickness.

*The mamelons worn out soon after eruption.

*The incisal ridges are inclined lingually to the root axis.

Mandibular central incisor

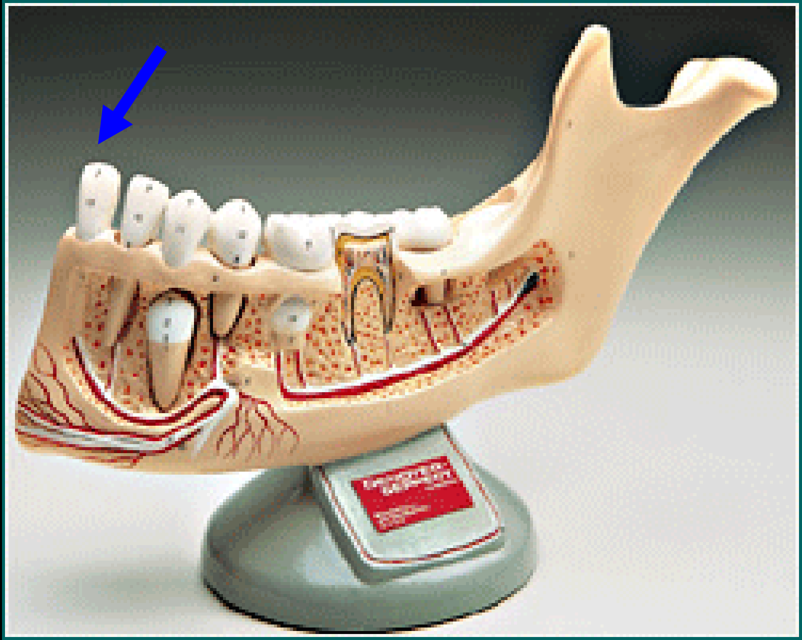
$\overline{2}$

Mandibular central incisor

Is the first mandibular tooth from the midline.

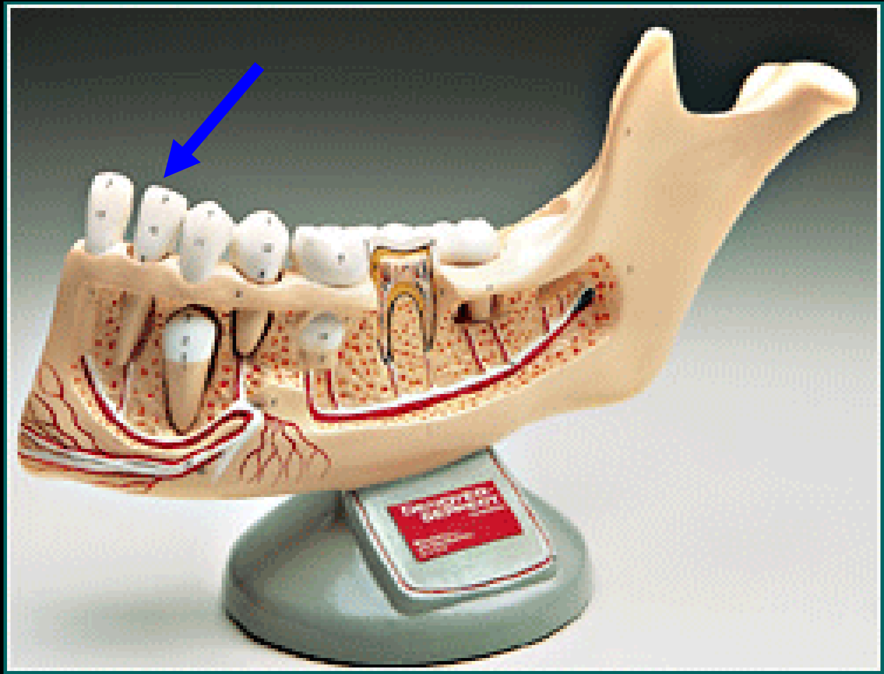
It is the smallest tooth in the permanent dentition.

It is the most symmetrical tooth in the permanent dentition.



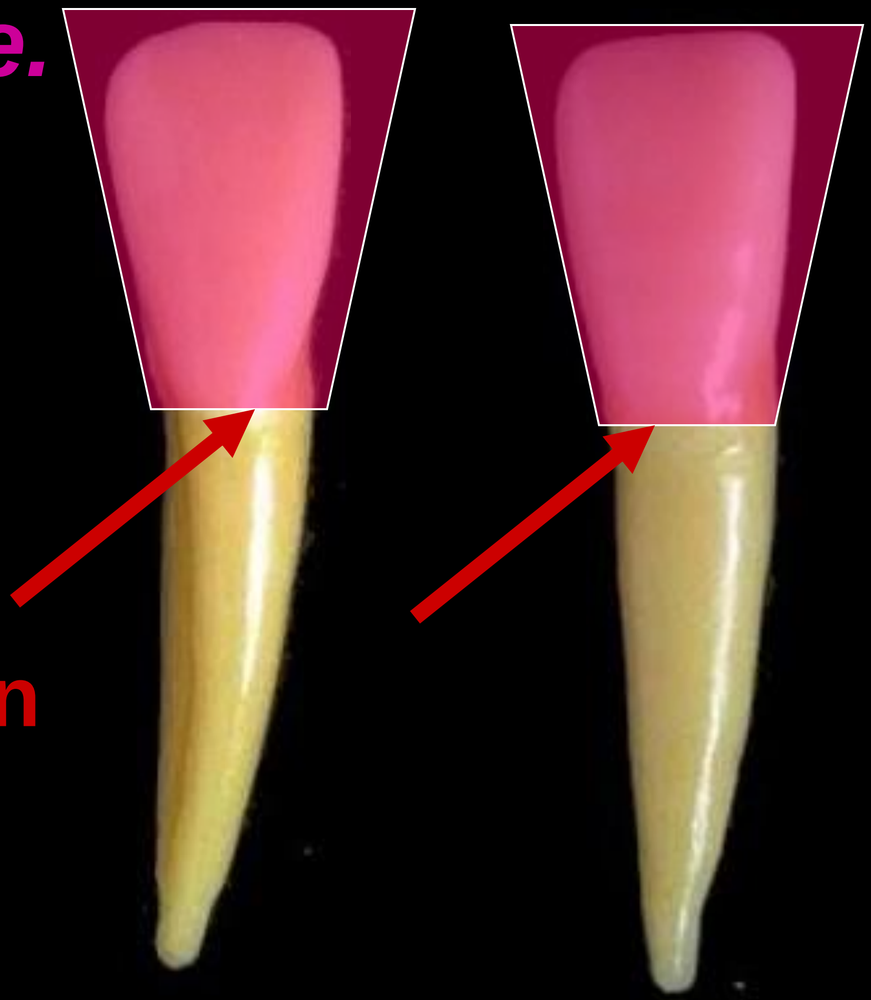
Mandibular lateral incisor

It’s very similar to lower central incisor, but slightly larger.



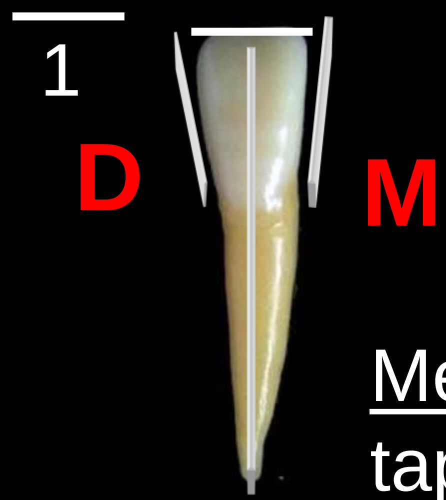
Geometric outline of the crown

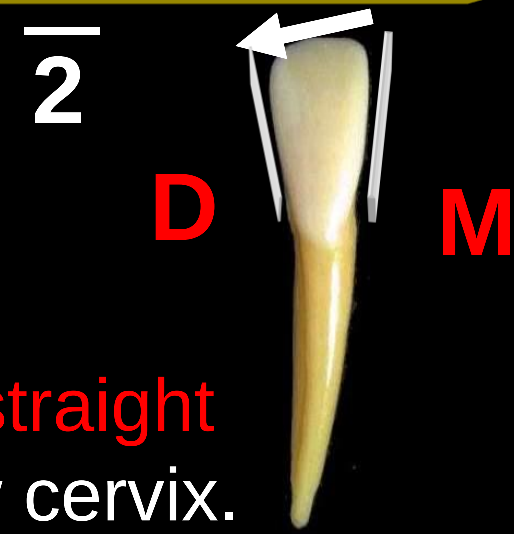
Labial and lingual surfaces have trapezoid outline.



The smallest uneven side cervically.

Labial and lingual outlines of the crown



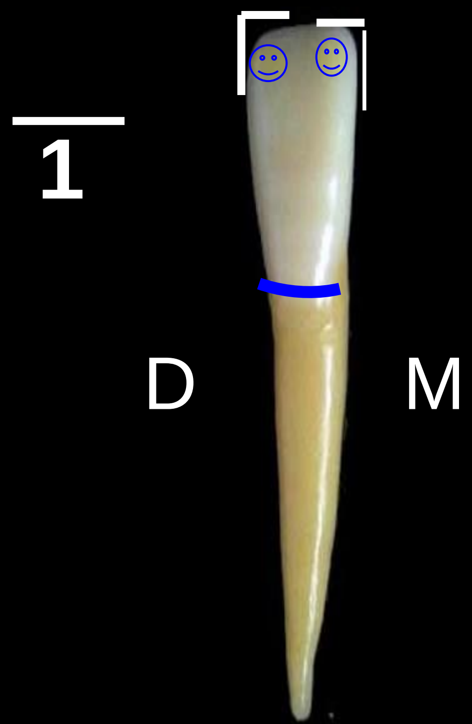


Mesial & distal outlines are straight tapering evenly to the narrow cervix.

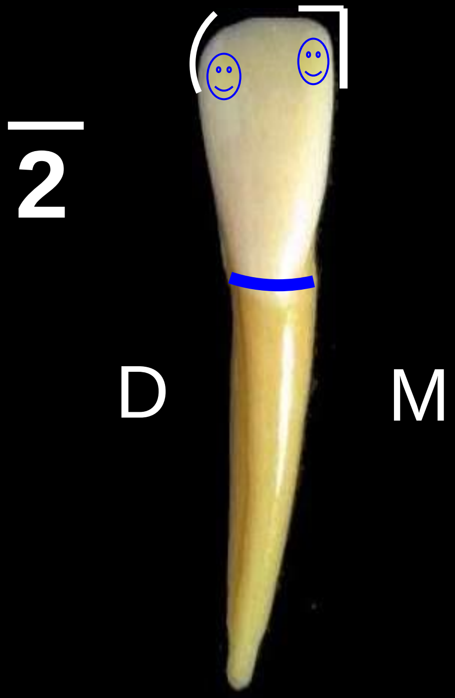
Incisal outline

is straight and perpendicular on the tooth long axis. Mamelons are present on newly erupted teeth.

The incisal ridge is inclined distally



Cervical outline is convex root wise.



Mesio and disto incisal angles are sharp.

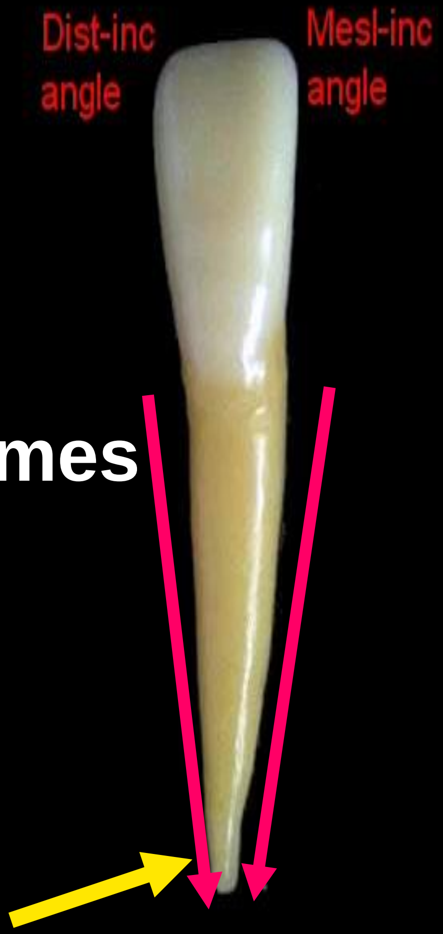
Mesio incisal angle is sharp while disto incisal angle is rounded.

Contact areas:

Mesially and distally are at the same level (I 1\3).

Mesially is at the I 1\3 while distally is more cervically.

Outlines of the root



- The mesial and distal outlines of the root are tapered to a pointed apex

- The apex is inclined distally but sometimes it’s straight.

Surface anatomy of the crown and root.

Labial surface of $\overline{1}$&$\overline{2}$

Elevations:

- **The crown surface is smooth and convex with maximum convexity at the cervical third (cervical ridge)**

Depressions:

•Shallow developmental grooves could be seen separating the mamelons in newly erupted teeth.

•The root surface is smooth and convex

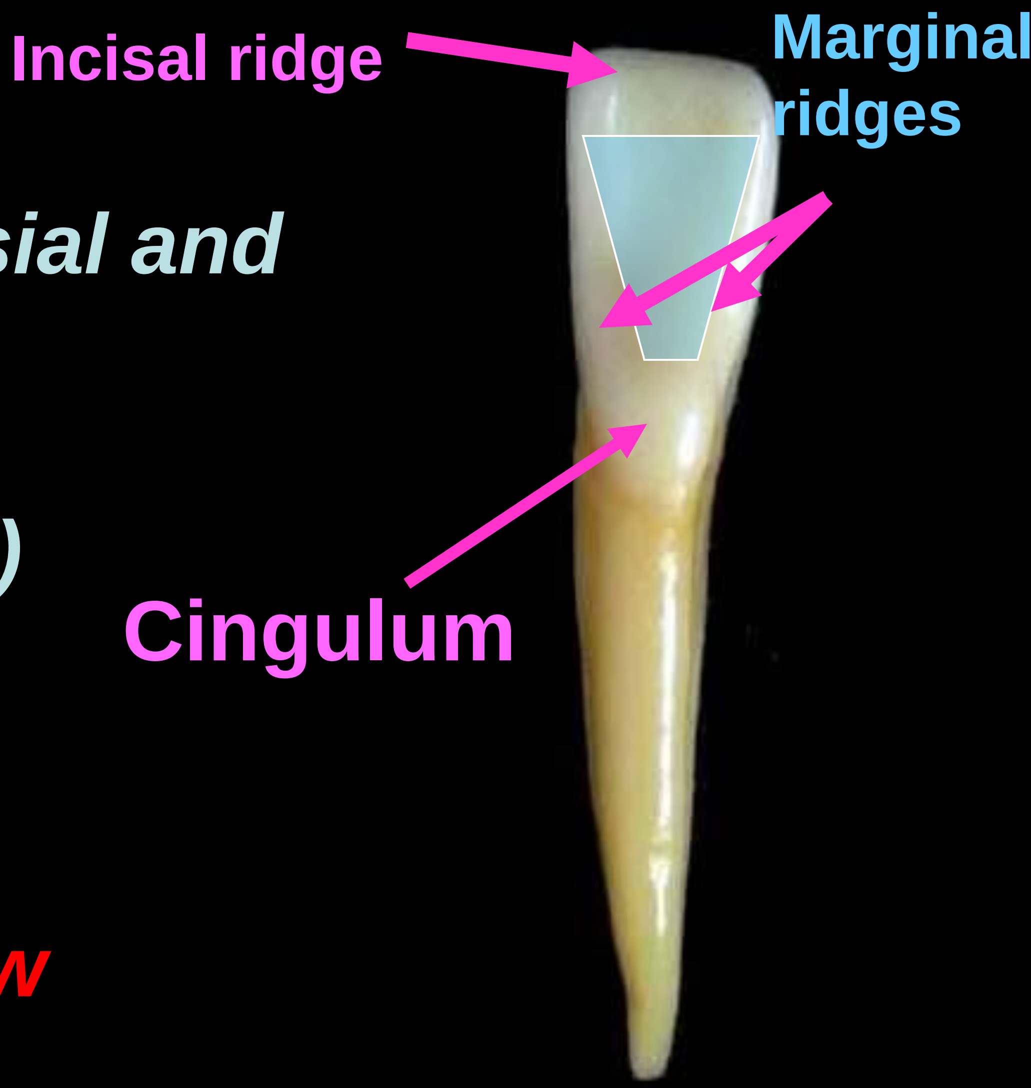
Lingual surface

Elevations:

- *Marginal ridges (mesial and distal)*
- *Cingulum (cervically)*
- *Incisal ridge*

Depressions:

Lingual fossa, shallow



Note:

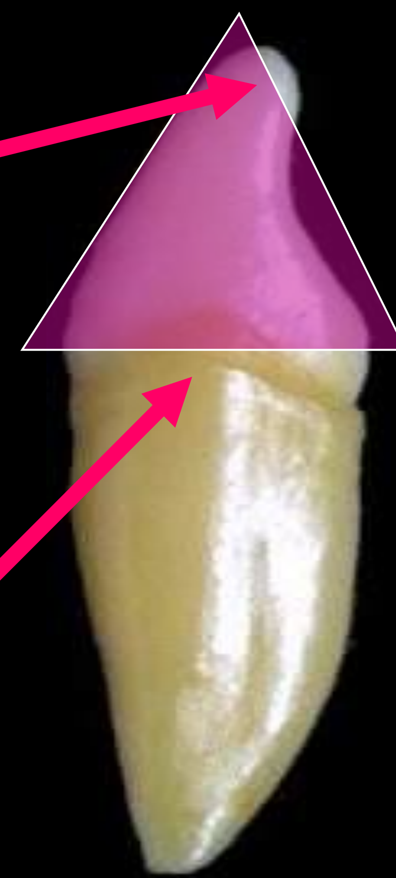
- ***All elevations are more developed in upper teeth than in lower teeth.***
- ***So the fossae appear shallower in the lower teeth.***
- ***In lower lateral the cingulum is shifted distally.***

Proximal (mesial &distal) surfaces

The geometric outline is triangular in shape

Apex incisally.

Base cervically.



The outline form of the proximal surfaces

It is formed of **:**

- ***Labial outline: convex with maximum convixity at the cervical third which represent …….***
- ***Lingual outline:***

convex incisally which represent…….

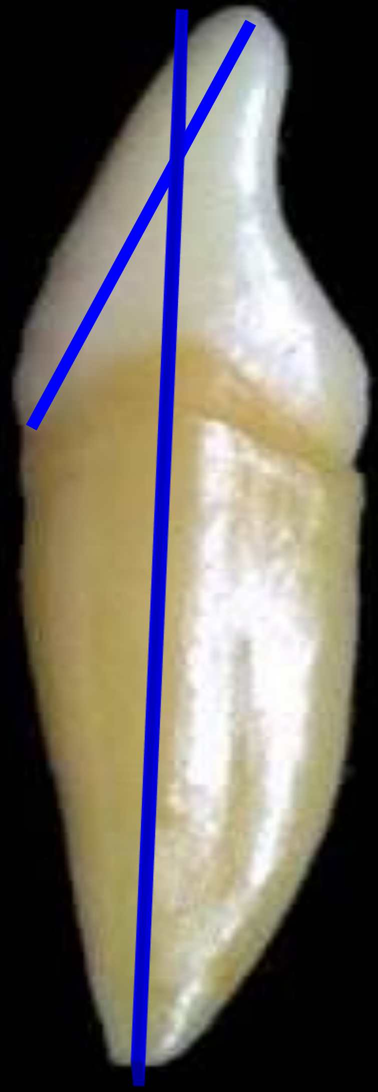
Concave at the middle which represent…

Convex cervically which represent………

- ***Cervical outline: Curves incisally.***

The incisal ridge is lingual to the root axis

This lingual inclination facilitates proper occlusion and also provides sufficient overlap and overjet to prevent lip biting .



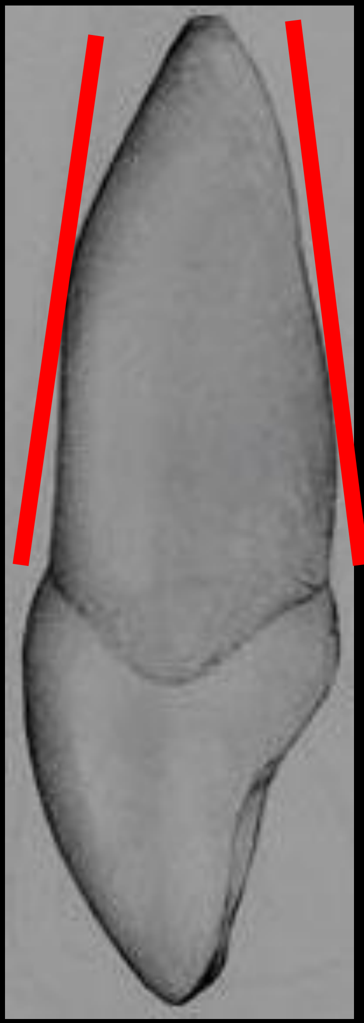
Outlines of the root

Upper incisors:

The outlines tapered from the cervical line to a blunt apex

Lower incisors:

The outlines are nearly straight & parallel from the cervical line to the middle third then tapered to a pointed apex



Surface anatomy of the crown and root.

***The crowns* have smooth convex proximal surfaces.**

Note: the contact areas mesially and distally are nearly at the same level but still the distal contact area is present more cervically.

1

2

Mesial surface

Contact area at the I 1\3

Distal surface

Contact area is more cervically to make contact with the lower canine

Note: the distal surface is shorter than mesially due to distal inclination of the incisal ridge

Roots

Root surface showed longitudinal developmental depression which is deeper distally than mesially.

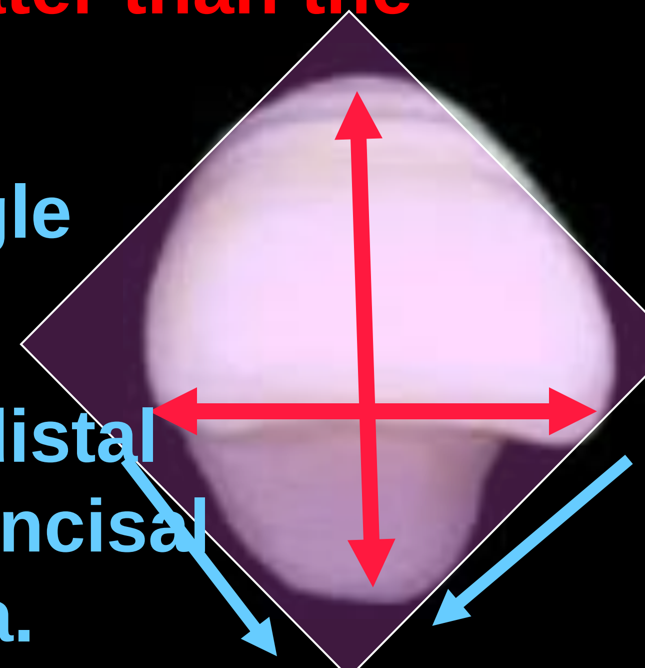
Incisal aspect
Outline and surface anatomy

The geometric outline is diamond in shape.

The labiolingual (thickness) is greater than the mesiodistal dimension.

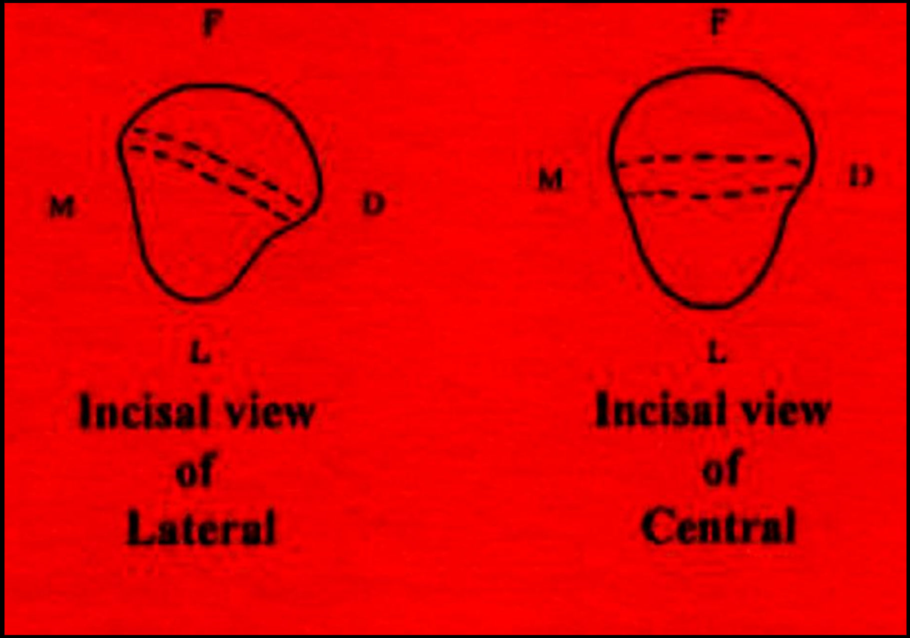
The incisal ridge in $\overline{1}$ is at right angle

to a line bisecting the crown labiolingually. Notice the mesial &distal marginal ridges,cingulum and the incisal ridge surrounding the lingual fossa.



The crown converges lingually.

*The incisal ridge of $\overline{2}$ is inclined lingually at it's distal end . This allows the tooth to follow the curvature of the dental arch .



Thank you